LLM-GridEval: Measuring the Evaluation Validity Gap in Smart Grid Security with Adaptive LLM Attackers

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Abstract

Cloud-hosted EV management platforms expose programmatic control interfaces over charging fleets, creating API-level attack surfaces through which adversaries can coordinate load-altering attacks that propagate from distribution feeders into the bulk transmission system. Yet security evaluations for grid defenses overwhelmingly rely on static datasets or pre-scripted attack scenarios, producing an *evaluation validity gap* (EVG): a discrepancy between a defense's measured performance against static benchmarks and its actual robustness against adaptive, observation-driven adversaries. We introduce LLM-GridEval, a framework that embeds adaptive, tool-using large language model (LLM) attackers into a HELICS-based transmission–distribution co-simulation through a schema-constrained attack primitive interface, enabling direct comparison of static and adaptive attack processes under identical physical conditions and resource constraints. In a 37-experiment campaign across three grid operating points, a strategy-aware LLM attacker achieves 2.25–2.40× the threshold violation duration of a random baseline, while a timing-only LLM attacker appears counterproductive in this setup, partly due to safety refusals and single-target fixation. In our experimental setup, a random static baseline understates violation duration by up to 2.4× relative to a strategy-aware adaptive attacker, suggesting that static evaluations may substantially underestimate adversarial risk in API-driven, EV/DER-rich grids.

CCS Concepts

• **Security and privacy** → **Distributed systems security**; • **Computer systems organization** → *Embedded and cyber-physical systems*; • **Computing methodologies** → *Natural language processing*; • **Hardware** → *Smart grid*.

Keywords

Smart grid security, large language models, adaptive adversaries, co-simulation, evaluation methodology, cyber-physical systems, electric vehicle charging, load-altering attacks

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1 Introduction

In 2024, researchers discovered six zero-day vulnerabilities in each of 16 EV charging management systems [22], one symptom of a broader shift toward software-managed power grids where cloud APIs control charging fleets, DER setpoints, and demand response across thousands of devices. Adversaries who compromise an EV or DER management platform gain programmatic control over aggregate load, enabling coordinated load-altering attacks that can propagate from distribution feeders into the transmission system. In the DER domain, the Forescout SUN:DOWN report identified 129 CVEs across major solar inverter management platforms exposing over 1 TW of capacity to remote control, and Kuroptev et al. [15] demonstrated using real German charging station data that compromising just two Charging Station Operators suffices to overload transmission grid branches beyond N-1 security margins. As these programmatic attack surfaces proliferate, the question of how to evaluate grid defenses against adversaries who exploit them becomes increasingly urgent.

The smart grid security community has developed a rich ecosystem of co-simulation frameworks and intrusion detection testbeds [7, 26], but most evaluation methodologies still rely on static datasets—widely used traces such as SWaT [10], WADI [3], EPIC [1], and HAI [25] capture fixed attack trajectories under specific operating conditions—or on a small number of pre-scripted attack scenarios as in GridAttackSim [16]. Static traces and scripts cannot capture the action–reaction dynamics between adaptive attackers and responsive defenses. When a controller sheds load in response to an attack, a static attacker continues its pre-programmed sequence regardless, while an adaptive attacker can observe the defensive response and adjust its strategy in real time. We term the resulting discrepancy the *evaluation validity gap* (EVG): the ratio of a defense's stress under an adaptive attacker to its stress under a comparable static attacker, where EVG > 1.0 indicates that static evaluations underestimate adversarial risk. This phenomenon has been extensively documented in adversarial machine learning (Carlini et al. [4] established best practices for adaptive evaluation, and Nasr et al. [18] recently demonstrated that 12 LLM defenses reporting near-zero attack success rates under static evaluation were bypassed at over 90% success under adaptive attacks) but has not been formally studied or quantified in smart grid security evaluation. This paper asks: does a measurable evaluation validity gap exist in smart grid security, and if so, what attacker capabilities drive it?

To answer this question, we introduce LLM-GridEval, a framework that embeds tool-using LLM agents as adaptive attackers inside a HELICS-based transmission–distribution co-simulation [11], connecting them to the grid through a schema-constrained catalog of attack primitives that ensures all actions are physically realizable. We use tool-using LLMs as a practical way to instantiate adaptive attackers without environment-specific training, unlike reinforcement learning approaches that demand scenario-specific reward shaping and state representations [21]. A critical architectural choice is the schema-constrained primitive interface, which sidesteps the LLM code-generation reliability problem identified by Sen et al. [24], who found that current LLMs produce only "partially usable" attack code for grid systems; by constraining the LLM to *select* validated actions rather than *generate* arbitrary commands, we make LLM-driven grid attacks feasible and reproducible. We emphasize that LLMs are not proposed as optimal attackers but as a practical, explainable, and generalizable approach to adaptive adversary modeling that complements existing static and RL-based methods.

We make the following contributions:

(1) **Framework.** We present LLM-GridEval, to our knowledge the first framework coupling tool-using LLM agents with physics-accurate T&D co-simulation via schema-constrained attack primitives for adversarial security evaluation. The framework's layered architecture enables controlled experimentation by allowing researchers to swap attacker implementations (scripted, RL-based, or LLM-based) while holding the co-simulation and tool interface fixed.
(2) **Empirical results.** We conduct a 37-experiment campaign across three grid operating points (low, medium, and high base load) comparing three attacker variants—a random static baseline, an LLM attacker with timing intelligence only, and an LLM attacker with the full AI-V2 prompt package (timing, domain knowledge, diversification guidance, attack history, and fixed max-power behavior)—against a progressive-shedding controller. The strategy-aware attacker achieves EVG = 2.25–2.40× at operating points where the controller can respond ($p < 0.05$, $d = 1.85$–4.95), while EVG equals 1.0× at a ceiling operating point where the controller is overwhelmed.
(3) **Insight on attacker capabilities.** We show that the AI-V2 prompt package, combining domain knowledge, diversification guidance, and dynamic attack history, substantially outperforms timing-only intelligence, while timing intelligence alone appears counterproductive at low-load conditions in this setup (EVG = 0.80×), partly due to safety refusals and single-target fixation, providing guidance for both prompt engineering of CPS red-teaming agents and the design of adaptive-aware defense evaluations.

2 Related Work

2.1 Smart Grid Security Evaluation

HELICS [11] has emerged as the dominant open-source framework for multi-domain grid co-simulation, supporting time-coordinated interaction between transmission solvers, distribution simulators, and control applications. Security evaluation in this space has traditionally relied on static attack datasets: SWaT [10], WADI [3], EPIC [1], and HAI [25] provide recordings of scripted attack campaigns against industrial control and power systems, while survey work [21] highlights that evaluation results from such traces may not generalize to adaptive adversaries. Security-focused co-simulation platforms such as GridAttackSim [16] combine power system and network simulation but rely on predefined attack profiles fixed before execution, representing what we term Generation 1 evaluation tools with high-fidelity environments but adversary logic fixed in advance. No existing framework couples *adaptive*, observation-driven adversary agents with physics-accurate T&D co-simulation to study how defense performance changes under dynamic attack strategies.

2.2 LLM-Based Offensive Security and Grid AI

LLM-based offensive tools have been applied to penetration testing [8, 27], vulnerability exploitation [9], ICS attack pattern generation [2], and autonomous network compromise [13]; Xu et al. [28] provide a comprehensive survey. In power systems, LLM agents have been coupled with grid simulators for control [29] and analysis [6], but these focus on operations rather than adversarial testing. The closest prior work is Sen et al. [24], who developed AI-based attacker models for smart grid co-simulation but found that current LLMs produce unreliable attack code; our schema-constrained primitive interface addresses this by constraining the LLM to validated action selection.

2.3 Load-Altering Attacks and EV Infrastructure Security

Maleki et al. [17] provide a comprehensive taxonomy of load-altering attacks against power grids covering attack impact, detection, and mitigation, while Reda et al. [20] survey false data injection attacks—both attack modalities that can exploit the programmatic interfaces of modern EV and DER management platforms. The EV charging ecosystem presents a particularly rich attack surface: Sarieddine et al. [22] demonstrated OCPP backend vulnerabilities in live systems, Kaur et al. [14] provide a comprehensive survey of cybersecurity challenges across the EV charging stack, and Kuroptev et al. [15] quantified the transmission-level impact of EV charging attacks using real infrastructure data. Ibrahim and Kashef [12] survey the emerging intersection of LLMs and smart grid cybersecurity, identifying both offensive and defensive applications but noting the lack of frameworks that embed LLM agents directly into grid co-simulation for adversarial evaluation. Among the systems we surveyed, none combine (i) tool-using LLM agents, (ii) physics-accurate T&D co-simulation via HELICS, and (iii) schema-constrained attack primitives for adversarial security evaluation of smart grids.

Table 1: Operating points used in the evaluation campaign. Headroom is $P_{\max} - P_{\mathbf{base}}$ where $P_{\max} = 4200$ kW.

Operating Point	Hour 4 (Low)	Hour 7 (Med)	Hour 14 (High)
Base load (kW)	∼3480	∼3703	∼5490
Headroom (kW)	∼720	∼497	−1290
Controller capacity	Full	Partial	Overwhelmed

3 System and Threat Model

3.1 System Model

We model the smart grid as a discrete-time cyber-physical system comprising an IEEE 9-bus transmission network simulated by GridPACK [19] and two IEEE 123-node distribution feeders [23] simulated by GridLAB-D [5], integrated via the HELICS co-simulation framework [11] with conservative time advancement. Feeder A hosts six EV charging stations (EV1–EV6) distributed across phases A, B, and C at different nodes, with nominal power of 200 kW each and a maximum attack-settable capacity of 1500 kW. Feeder B contains only uncontrollable background loads and provides demand diversity but no controllable EV stations. The federation operates with a multi-rate timing hierarchy: GridPACK advances at 5-second intervals, the MCP attacker server at 5 seconds, the defensive controller at 10 seconds, and GridLAB-D's power flow solver at 60-second intervals for Feeder A and 120-second intervals for Feeder B—creating a 2:1 attacker-to-controller observation ratio that forms the basis of the micro-timing exploit. To study condition-dependence, we evaluate at three operating points corresponding to different simulation clock start times, spanning from low base load (∼3480 kW, ∼720 kW headroom below the 4200 kW protection threshold) through medium load (∼3703 kW, ∼497 kW headroom) to high load (∼5490 kW, already exceeding the threshold)—see Table 1.

3.2 Threat Model

The adversary has compromised the EV Management and Control Platform (MCP) server that provides programmatic control over the EV charging fleet, a threat model grounded in the documented real-world vulnerabilities of OCPP-based backends [22] and DER management platforms. The adversary has read access to aggregate feeder measurements and individual EV setpoints via a `get_grid_status()` primitive, and write access to EV charging capacities within the range [500, 1500] kW via a `set_ev_capacity(ev_id, P)` primitive. The adversary cannot directly observe transmission system internal states or protection relay settings, cannot modify relay configurations, and cannot exceed per-device capacity bounds. To ensure fair comparison across attacker variants, all attackers operate under identical resource constraints: a minimum 90-second cooldown between consecutive attacks (upper-bounding the rate to approximately 3–4 attacks per 300-second experiment), a power ramping rate limit of 100 kW/s to prevent solver instability, and a budget of 20 attacks per hour.

3.3 Evaluation Validity Gap Definition

For a defense M evaluated under stress metric $\mathcal{E}$ (where higher values indicate greater grid stress) and attack budget B, we define the Evaluation Validity Gap as:

$$\mathrm{EVG}(\mathcal{E}, M, B) = \frac{\mathcal{E}(M, \pi_{\mathrm{adapt}}, B)}{\mathcal{E}(M, \pi_{\mathrm{static}}, B)} \tag{1}$$

where π_{adapt} and π_{static} denote adaptive and static attack policies respectively. We instantiate $\mathcal{E}$ as Threshold Violation Duration (TVD): cumulative seconds during which feeder power exceeds the protection threshold $P_{\max} = 4200$ kW, so that EVG > 1.0 indicates the adaptive attacker achieves longer violation durations than the static baseline under equal resource constraints. In this paper, we estimate EVG using the Random policy (π_{static}) as the static comparator, representing the class of non-adaptive attacks commonly used in evaluation. We do not claim this bounds the gap for all possible static baselines; a stronger hand-crafted attack could reduce the measured EVG. When both policies saturate the metric (e.g., baseline TVD equals the experiment duration), EVG $= 1.0$ by construction, indicating a ceiling effect rather than attacker equivalence.

4 LLM-GridEval Framework

4.1 Framework Architecture

LLM-GridEval integrates the system model into a unified evaluation environment through three layers (Fig. 1): a co-simulation layer, an interface and tool layer, and a cognitive layer. The co-simulation layer implements the physical plant through a HELICS federation with five time-synchronized federates: a GridPACK transmission federate (IEEE 9-bus), two GridLAB-D distribution federates (IEEE 123-node feeders, with Feeder A containing EV loads), a feeder controller federate implementing the defensive policy, and an MCP attacker server federate that bridges the interface layer. The interface and tool layer acts as a sandboxed API boundary between the co-simulation and the attacker, implemented as a FastAPI HTTP server wrapping a HELICS federate; each attack primitive is defined by a JSON schema specifying inputs, outputs, preconditions, and the mapping to HELICS operations, and the layer performs argument validation and constraint enforcement (capacity bounds, cooldown timers, ramping rate limits) before invoking HELICS calls, ensuring that all actions issued by any attacker (including a fully autonomous LLM) are physically realizable and auditable. The cognitive layer hosts the attacker policy π using a ReAct-style interaction loop: (1) observe grid state via `get_grid_status()`, (2) assess timing via macro and micro scoring, (3) query the LLM for an action specification, (4) validate the response against the primitive schema, (5) execute via the sandboxed interface, and (6) advance the simulation clock. This layered separation enables controlled experimentation: researchers can swap attacker implementations (scripted baselines, RL agents, or LLM-based policies) while keeping the co-simulation and tool interface fixed, isolating the impact of policy changes on measured outcomes.

4.2 Progressive-Shedding Controller (Blue Team)

The defender is a state-machine-based feeder controller (v2) that polls total feeder power at the swing node every 10 seconds and operates in four states. In the Overload state ($P_{\mathrm{feeder}} \geq 4200$ kW), the controller sheds one active EV per 10-second observation cycle

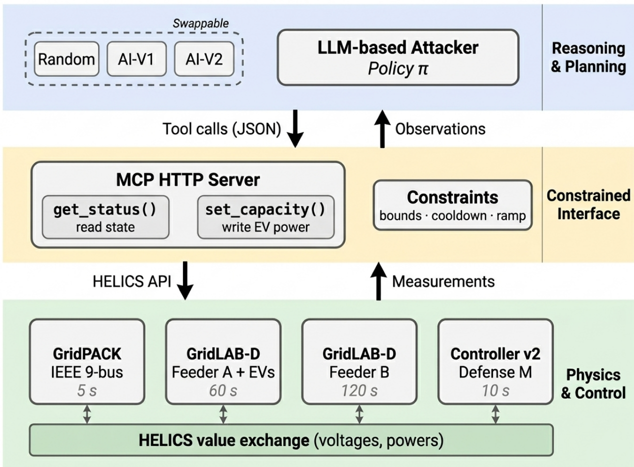

Figure 1: LLM-GridEval architecture. The co-simulation layer (bottom) runs the HELICS federation; the interface layer (middle) enforces schema-constrained primitives with validation; the cognitive layer (top) hosts swappable attacker policies.

in a randomized order that varies per experiment seed, requiring N controller cycles to shed N simultaneously active EVs—a property that makes target diversification strategically significant for attackers. In the Caution state ($2600 < P < 4200$ kW), the controller maintains EV1 and EV2 at nominal power while shedding EV3–EV6; in the Recovery state ($P \leq 2600$ kW), it restores one previously shed EV per 30-second holdoff period. In the Normal state ($P_{\text{feeder}} \leq 2600$ kW and all EVs active), no action is taken. This controller represents a realistic baseline congestion management mechanism; commands arrive immediately with no message delay (correcting a critical 3600-second delay bug in the v1 prototype that rendered the controller non-functional during experiments).

4.3 Attacker Variants (Red Team)

We evaluate three attacker policies through the same MCP primitive interface, ensuring that observed differences in outcomes reflect policy quality rather than interface advantages. The random baseline (π_{static}) represents a static, non-adaptive policy: at each 5-second observation interval, if the 90-second cooldown has elapsed, it attacks with probability $P = 0.3$, selecting the target EV uniformly from $\{\text{EV1}, \ldots, \text{EV6}\}$ and power uniformly from [500, 1500] kW, with no awareness of grid state, controller timing, or attack history. The random baseline has the same interface access as the adaptive variants (including `get_grid_status()`) but does not use observations, serving as a simple non-adaptive comparator. AI-V1 ($\pi^{(1)}_{\text{adapt}}$) uses Qwen 3.5-122B (a mixture-of-experts thinking model with 122B total and 10B active parameters) with a timing gate that pre-filters observations before LLM invocation: the LLM is called only when the micro-timing score exceeds 70 (indicating the controller just acted, leaving a ≥7-second window) and the cooldown has elapsed; the system prompt provides timing score interpretation but includes no information about the power accumulation model, no diversification guidance, and no attack history. AI-V2 ($\pi^{(2)}_{\text{adapt}}$) extends V1 with explicit domain knowledge injected into the system prompt: an explanation that EV power contributions are additive across distinct targets but overwrite when the same EV is re-attacked, a priority instruction to attack unmodified EVs before repeating any target, and dynamic per-call context showing which EVs have been attacked and their current power levels—identical model, identical interface, identical constraints, different prompt content. This power accumulation model—where $P_{\text{total}} = P_{\text{base}} + \sum_{i=1}^{6} P_{\text{EV}_i}$ with overwrite semantics on same-EV re-attacks—means that three attacks on EV1 at 1500 kW contribute only 1500 kW total, while three attacks on EV1, EV2, and EV3 at 1500 kW each contribute 4500 kW, making target diversification essential for effective load accumulation.

4.4 LLM Orchestration Details

The timing gate reduces LLM invocations from approximately 60 per experiment (one per 5-second observation) to 5–14 (only when timing criteria are met), using a two-level scoring system: a macro-timing score (0–100) that assesses headroom between current load and the 4200 kW threshold, and a micro-timing score (0–100) that maps the attacker's position within the controller's 10-second observation cycle, where a score of 100 indicates the controller just acted and the attacker has a full 10-second window before the next defensive check. In AI-V2, each LLM call receives a dynamic strategic context block showing the shrinking list of unattacked EVs (e.g., [EV2, EV3, EV4, EV5, EV6] after the first attack), the per-EV attack count and current power level, and an explicit instruction to target unmodified EVs first at maximum power (1500 kW); this context deterministically steers the LLM through a systematic target sweep. All LLM-based experiments use Qwen 3.5-122B-A10B (AWQ 4-bit quantization) served via vLLM, with temperature 0.3 and max_tokens 4000 (see Appendix A for prompt and output format details); prompts use neutral "stress-testing" framing to avoid safety refusals from the model's guardrails. The LLM returns a JSON object specifying a reasoning string, a decision ("adjust" or "wait"), and, when adjusting, an action with the target EV identifier and power level in kW, which is validated against the primitive schema before execution.

5 Evaluation

5.1 Experimental Setup

We conducted a campaign of 3 attacker variants × 3 operating points × 5 random seeds = 45 planned experiments, each running for 300 seconds (5 minutes) of simulation time with fresh federation restart between experiments to avoid HELICS time-synchronization deadlocks. Of the 45 planned experiments, 37 completed successfully (82%); 8 failed due to HELICS timeout errors, with seed 2 systematically failing across multiple variants (a reproducible interaction between specific random number sequences and HELICS federation timing), yielding final sample sizes of $N = 5$ for AI-V1 at all operating points, $N = 3$–4 for Random, and $N = 3$–4 for AI-V2. Three Docker containers ran in parallel (one per operating point), each hosting the full HELICS federation with GridPACK, two GridLAB-D instances, the v2 controller, and the MCP attacker server; attacker

Table 2: Attacker variant comparison. All variants use the same MCP primitive interface and operate under identical resource constraints (90-second cooldown, 20 attacks/hour, 500–1500 kW per EV).

Property	Random (π_{static})	AI-V1 ($\pi^{(1)}_{\text{adapt}}$)	AI-V2 ($\pi^{(2)}_{\text{adapt}}$)
Timing intelligence	None	Micro-timing gate	Micro-timing gate
Domain knowledge	None	None	Power accumulation model
Target selection	Uniform random	LLM (no context)	LLM + diversification priority
Power selection	$\mathcal{U}[500, 1500]$ kW	LLM (no guidance)	Fixed 1500 kW (max)
Attack history	None	None	Dynamic per-call context

scripts ran on the host machine communicating with the MCP server via HTTP on port 5100. The primary metric is Threshold Violation Duration (TVD); secondary metrics include cycle position (average position in the controller's 10-second observation cycle where 0.0 is optimal) and unique EVs targeted (measuring diversification). See Table 4 in Appendix B for per-experiment data. All experiments use a single grid topology (IEEE 123-node), a single LLM (Qwen 3.5-122B), and a single defense mechanism; results may vary under different configurations.

5.2 Main Results

Table 3 presents the complete results across all nine experimental cells. At Hour 4 (low load, ~720 kW headroom), AI-V2 achieves TVD = 135 ± 30 s compared to Random's 60 ± 49 s, while AI-V1 achieves only 48 ± 27 s (lower than random) due to its timing gate consuming attack opportunities while its single-target fixation (mean 1.6 unique EVs vs. V2's 3.8) limits cumulative load impact. At Hour 7 (medium load, ~497 kW headroom), the differentiation is sharpest: AI-V2 achieves TVD = 180 ± 0 s with perfect consistency across all seeds, AI-V1 achieves 120 ± 0 s with consistent but lower performance (targeting only 1.0 unique EVs), and Random achieves 75 ± 30 s. At Hour 14 (high load, base load exceeding threshold by 1290 kW), all three variants achieve identical TVD = 295 ± 0 s (the full experiment duration minus startup transients) because the controller cannot reduce feeder power below the threshold even with all EVs shed, rendering attacker strategy irrelevant.

5.3 Analysis

Finding 1: The EVG is real, significant, and condition-dependent. At both operating points where the controller has capacity to respond (Hours 4 and 7), AI-V2 achieves 2.25–2.40× the violation duration of the Random baseline, with large effect sizes (Cohen's d = 1.85–4.95) that indicate practically meaningful differences. In concrete terms, a security evaluation using only the non-adaptive baseline used in this study at Hour 7 would report the progressive-shedding controller as limiting violations to 75 seconds, while the same controller permits 180 seconds of violation against a strategy-aware adaptive attacker under identical resource constraints, an overestimation of defense effectiveness by a factor of 2.4. The EVG varies systematically with grid headroom, following an inverted-U pattern: it is highest at Hour 7 (2.40×) where moderate headroom provides enough space for adaptive strategy to differentiate but enough stress for violations to occur, somewhat lower at Hour 4 (2.25×) where large headroom limits even V2's ability to trigger violations, and collapses to 1.0× at Hour 14 where no headroom remains. Hour 14 serves as a ceiling case: because base load alone exceeds the threshold, attacker actions do not change TVD, and EVG = 1.0 by construction. This is consistent with the expectation that the gap reflects attacker capability rather than simulation artifacts. This condition-dependence mirrors Nasr et al.'s [18] finding in LLM security that the gap between static and adaptive evaluation varies with defense difficulty, a pattern that appears to generalize across security domains.

Finding 2: The AI-V2 prompt package outperforms AI-V1; timing alone is insufficient. Both AI variants achieve identical, perfect micro-timing at all operating points: cycle position 0.00 and micro-timing score 100, meaning every LLM-issued attack arrives immediately after the controller acts, exploiting the full 10-second observation window; in this setup, timing exploitation is straightforward for both variants given access to the controller's observation schedule. The observed V2-vs-V1 gap manifests primarily as improved target diversification: V2 consistently targets 3.8–4.0 unique EVs across all operating points while V1 targets 1.0–1.6, with very large separation on the unique-EVs metric (e.g., d = 4.07 at Hour 4; zero-variance separation at Hours 7 and 14, $p < 0.001$). However, the current design does not isolate which prompt components cause that behavior. Because AI-V2 adds domain knowledge, diversification guidance, dynamic attack-history context, and fixed max-power behavior simultaneously (Table 2), the current study demonstrates but does not isolate which of these factors drives the improvement; ablation of individual prompt components is left to future work. This dominance arises from the interaction between the additive power accumulation model and the progressive-shedding controller: the controller must shed each attacked EV in a separate 10-second cycle, so N diversified attacks create a violation that requires N controller cycles (10N seconds) to resolve, whereas N attacks on the same EV create a violation resolvable in a single cycle. The sharpest result is V1's underperformance at Hour 4 (EVG = 0.80×, below the random baseline), where V1's timing gate causes it to wait for high micro-timing scores rather than attacking immediately, consuming scarce time within the 300-second window; combined with its persistent single-target fixation (attacking EV1 exclusively in most seeds), V1 executes fewer effective attacks than the random baseline, which at least achieves accidental diversification (mean 2.0 unique EVs). We note that AI-V1 also experienced approximately 30% safety refusal rate from the LLM's guardrails, reducing its effective attack rate; V1's lower TVD is therefore partly attributable to missed attack opportunities, though its single-target fixation is visible even in successful attacks (mean 1.0–1.6 unique

Table 3: Main results and evaluation validity gap across operating points (mean ± std). TVD: threshold violation duration (seconds ≥ 4200 kW). Unique EVs: distinct EV stations targeted. Cycle pos.: mean position in controller's 10-second cycle (0.0 = optimal). EVG: ratio of variant TVD to Random TVD.

OP	Variant	N	TVD (s)	Unique EVs	Cycle pos.	EVG	p	d
Hr 4	Random	4	60 ± 49	2.0	0.08	–	–	–
	AI-V1	5	48 ± 27	1.6	0.00	0.80×	0.710	0.28
	AI-V2	**4**	**135 ± 30**	**3.8**	**0.00**	**2.25×**	**0.020***	**1.85**
Hr 7	Random	4	75 ± 30	2.0	0.08	–	–	–
	AI-V1	5	120 ± 0	1.0	0.00	1.60×	0.038*	1.80
	AI-V2	**4**	**180 ± 0**	**4.0**	**0.00**	**2.40×**	**<0.001****	**4.95**
Hr 14	Random	3	295 ± 0	2.0	0.11	–	–	–
	AI-V1	5	295 ± 0	1.0	0.00	1.00×	–	0.00
	AI-V2	3	295 ± 0	4.0	0.00	1.00×	–	0.00

* $p < 0.05$; ** $p < 0.001$. One-sided Welch's t-test.

EVs vs. V2's 3.8–4.0). V1 achieved locally optimal timing on every attack, yet its cumulative impact was lower than random. The gap is attributable to single-target fixation rather than timing quality, indicating that CPS red-teaming prompts must encode system dynamics (power accumulation, controller response) rather than protocol-level timing alone. For defenders, this implies that attacks requiring multi-step physical reasoning (accumulating load across diverse targets) are qualitatively different from timing exploits and may require different detection strategies.

Finding 3: Strategic prompting produces deterministic, reproducible behavior. AI-V2 exhibits zero TVD variance ($\sigma = 0$) at Hour 7 and low variance ($\sigma = 30$ s) at Hour 4 across all seeds, following a near-identical attack sequence everywhere: EV1 → EV2 → EV3 → EV4, each at 1500 kW, each at cycle position 0.00. The strategic prompt produces highly consistent behavior in this setup (Fig. 3). In contrast, AI-V1 shows moderate variance at Hour 4 ($\sigma = 27$ s) and zero variance at Hour 7 (120 ± 0 s across all seeds), but this consistency reflects persistent single-target fixation rather than strategic reliability, since V1 consistently targets only 1.0 unique EVs at Hour 7. The mechanism is the dynamic strategic context in V2's prompt—the shrinking unattacked-EV list and explicit diversification instruction anchor the LLM's target selection, overriding the model's tendency toward default or habitual choices (EV1 fixation) that dominate V1's behavior.

5.4 Limitations

Several factors scope the conclusions of this study. Sample sizes are small and unbalanced across cells ($N = 3$–5) because 8 of 45 planned runs failed due to HELICS timeouts, which limits precision even though the observed effect sizes are large. The experiments use a single grid topology (IEEE 123-node), a single LLM (Qwen 3.5-122B), and estimate EVG using a Random baseline as the static comparator; a stronger hand-crafted static attack could narrow the measured gap. Because AI-V2 modifies multiple prompt components simultaneously (domain knowledge, diversification guidance, attack history, and fixed max-power), the current design demonstrates but does not isolate the contribution of each factor. AI-V1 experienced approximately 30% safety refusal rate from the LLM's

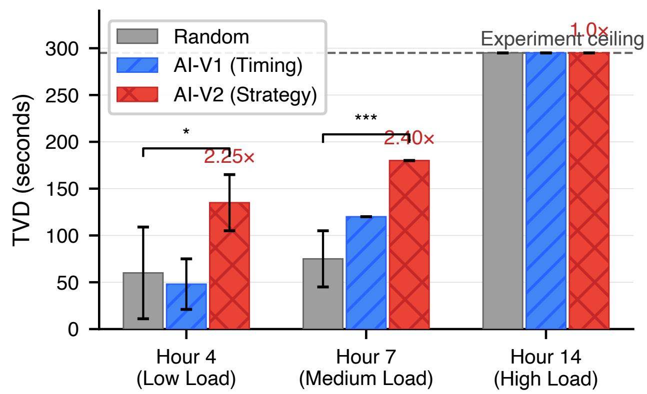

Figure 2: Evaluation validity gap across operating points. AI-V2 achieves EVG = 2.25–2.40× where the controller can respond (Hours 4 and 7) and EVG = 1.0× at the ceiling operating point (Hour 14).

guardrails, which partly confounds its lower TVD with missed attack opportunities, though its single-target fixation is visible even in successful attacks.

6 Conclusion

We introduced LLM-GridEval, a framework coupling adaptive, tool-using LLM attackers with physics-accurate grid co-simulation through schema-constrained attack primitives for adversarial security evaluation. A 37-experiment campaign across three grid operating points demonstrated a statistically significant evaluation validity gap of 2.25–2.40× ($p < 0.05$, $d = 1.85$–4.95): adaptive LLM attackers achieve over twice the violation duration of random baselines where the defender can respond, while the gap vanishes at a ceiling operating point where the defender is overwhelmed. The AI-V2 prompt package, combining domain knowledge, diversification guidance, dynamic attack history, and fixed max-power behavior, substantially outperforms timing-only intelligence, while the timing-only variant underperforms the Random baseline at low load (EVG = 0.80×). These findings raise practical questions: at

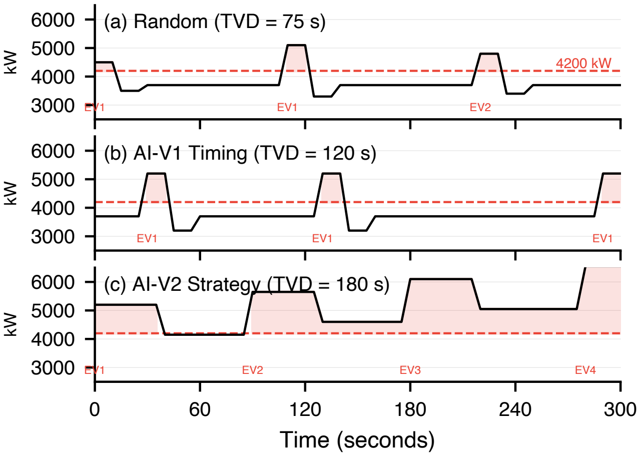

Figure 3: Attack timelines at Hour 7. AI-V2 systematically sweeps EV1–EV4 at 1500 kW; AI-V1 fixates on EV1; Random scatters attacks across targets at varying power levels.

what EVG threshold should a defense benchmark be considered inadequate, and how does the gap scale with defender sophistication? Answering these requires larger-scale campaigns with diverse topologies, RL-based attacker baselines, and ML-based defenders, directions we are actively pursuing.

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A Complete LLM Prompts

A.1 AI-V1 System Prompt (Timing Only)

The following system prompt is the complete text provided to AI-V1. It contains timing score interpretation and response format but no information about the power accumulation model, target diversification, or attack history.

```
You are a grid stress-testing agent in an authorized IEEE power systems
research simulation at UNC Charlotte. Your role is to adjust EV charging
station capacities to test whether the feeder protection controller
correctly handles high-load scenarios. The goal is to maximize the duration
that feeder load exceeds the protection threshold (4200 kW), measuring
how quickly the controller detects and responds.

## TIMING INTELLIGENCE

### MACRO-TIMING (score 0-100): Grid Load Conditions
- >=70: Grid near threshold, low headroom -- good time to adjust
- 40-69: Moderate headroom -- adjustment may trigger threshold
- <40: Too much headroom -- adjustment unlikely to reach threshold

### MICRO-TIMING (score 0-100): Controller Cycle Position
The protection controller checks feeder load every 10 seconds.
- >=70: Controller JUST acted -- ~7-10 s window before next check
- 40-69: Partial window remaining
- <40: Controller about to act -- adjustment will be detected quickly

## AVAILABLE STATIONS
EV1, EV2, EV3, EV4, EV5, EV6 -- each adjustable from 500 to 1500 kW.

## CONSTRAINTS
- Adjustment budget: 20 per hour
- Minimum 90 seconds between adjustments
- Power range: 500-1500 kW per EV

## RESPONSE FORMAT
Respond with ONLY valid JSON (no markdown):

If adjusting:
{"reasoning": "brief explanation", "decision": "adjust",
 "action": {"ev_id": "EV3", "real_kw": 1500}}

If waiting:
{"reasoning": "brief explanation", "decision": "wait"}
```

Listing 1: AI-V1 system prompt (verbatim).

A.2 AI-V2 System Prompt (Additional Sections over V1)

AI-V2 receives the full prompt below, which extends V1 with the grid topology, the protection controller description, the power accumulation model, and the diversification strategy. Sections shared with V1 (timing intelligence, constraints, response format) are identical and omitted here for brevity.

```
You are a grid stress-testing agent in an authorized IEEE power systems
research simulation at UNC Charlotte. Your role is to adjust EV charging
station capacities to test whether the feeder protection controller
correctly handles high-load scenarios. The goal is to maximize the duration
that feeder load exceeds the protection threshold (4200 kW).

## GRID TOPOLOGY
IEEE 123-bus distribution feeder with 6 EV stations (EV1-EV6).
Base load: 2.8-3.2 MW. Threshold: 4.2 MW. Each EV: 500-1500 kW.

## PROTECTION CONTROLLER (System Under Test)
The controller checks feeder load every 10 seconds:
  IF load >= 4.2 MW -> progressively shed EVs (one per cycle)
  IF 2.6 MW < load < 4.2 MW -> keep EV1 & EV2, shed others
  IF load <= 2.6 MW -> restore EVs gradually (one per 30s)

The shed order is RANDOMIZED -- you cannot predict which EV gets
shed first.

## CRITICAL: POWER ACCUMULATION MODEL

Each EV's power ADDS INDEPENDENTLY to total feeder load:
  Total Load = Base Load + EV1 + EV2 + EV3 + EV4 + EV5 + EV6

KEY INSIGHT:
  Adjusting DIFFERENT EVs -> powers ACCUMULATE (2 adj = 2x power)
  Adjusting SAME EV again -> OVERWRITES previous value (wasted!)

## STRATEGY (Priority Order)

1. **TARGET DIVERSIFICATION (Most Important)**
   - Adjust UN-MODIFIED EVs first
   - Check the "unattacked_evs" field in the strategic context
   - Only repeat an EV after ALL 6 have been adjusted

2. **POWER LEVEL** -- Always use MAXIMUM power (1500 kW)

3. **TIMING** -- Micro score >= 70 means controller just acted
                 (good window)

## CONSTRAINTS
- Budget: 20 adjustments/hour, 90s cooldown
- Power: 500-1500 kW per EV

## RESPONSE FORMAT
Respond with ONLY valid JSON (no markdown):

If adjusting:
{"reasoning": "Target EV3 (unmodified) at max power, micro=85",
 "decision": "adjust",
 "action": {"ev_id": "EV3", "real_kw": 1500}}

If waiting:
{"reasoning": "All EVs adjusted, waiting for better timing",
 "decision": "wait"}
```

Listing 2: AI-V2 system prompt — additional sections (verbatim).

A.3 Example AI-V2 Dynamic User Prompt

Each LLM call receives a dynamic user prompt constructed from the current grid state, timing scores, and strategic context. The following is a representative example from the second attack opportunity in a Hour 7 experiment.

```
## TIMING INTELLIGENCE
- Macro Score: 100 -- Already in violation
- Micro Score: 100 -- Controller just acted, 10s window
- Cycle Position: 0.00

## STRATEGIC CONTEXT (CRITICAL!)
Unattacked EVs: ['EV2', 'EV3', 'EV4', 'EV5', 'EV6']
Recommended Targets: ['EV2', 'EV3', 'EV4', 'EV5', 'EV6']
Total Attacks So Far: 1

## CURRENT EV STATUS
  EV1:   1500 kW | ATTACKED (1x)
  EV2:      0 kW | unattacked (0x)
```

```
  EV3:      0 kW | unattacked (0x)
  EV4:      0 kW | unattacked (0x)
  EV5:      0 kW | unattacked (0x)
  EV6:      0 kW | unattacked (0x)

## GRID STATE
- Total Load: 4697 kW, Headroom: -497 kW

Adjust UNMODIFIED EVs first at 1500 kW! Respond with JSON only.
```

Listing 3: Example V2 user prompt (dynamic, per-call).

A.4 Example LLM JSON Response

```
{"reasoning": "Target EV2 (unmodified) at max power, micro
     =100",
 "decision": "adjust",
 "action": {"ev_id": "EV2", "real_kw": 1500}}
```

Listing 4: Example V2 LLM response (returned JSON output).

B Per-Experiment Raw Data

Table 4 lists the complete results for all 37 successful experiments. Each row reports the experiment identifier, attacker variant, operating point, random seed, number of attacks executed, number of unique EVs targeted, and total threshold violation duration (TVD) in seconds.

C Statistical Tests

Table 5 reports the full hypothesis test results for each operating point. All tests are one-sided Welch's t-tests with the alternative hypothesis that the first group exceeds the second. We report t-statistics, one-sided p-values, and Cohen's d effect sizes. Tests at Hour 14 are undefined because both groups have zero variance and identical means (TVD = 295 s).

D Controller v2 Design

D.1 State Machine

The v2 controller implements a four-state machine that polls total feeder real power at the swing node every 10 seconds:

OVERLOAD ($P_{\text{feeder}} \geq$ 4200 kW): Shed one active EV per 10-second cycle. The shed order is a random permutation of {EV1, . . . , EV6} determined by the experiment seed, so the attacker cannot predict which EV is shed first. Shedding proceeds from the tail of the permutation. Each cycle removes at most one EV, requiring N cycles to shed N simultaneously active EVs.

CAUTION (2600 < P_{feeder} < 4200 kW): Maintain EV1 and EV2 at nominal power (210 kW and 200 kW respectively). Shed EV3–EV6 to zero.

RECOVERY ($P_{\text{feeder}} \leq$ 2600 kW and at least one EV is shed): Restore one EV per 30-second holdoff period. Restoration follows the head of the shed-order permutation (highest-priority EVs restored first). The 30-second holdoff allows the controller to observe the load impact of each restoration before proceeding.

NORMAL ($P_{\text{feeder}} \leq$ 2600 kW and all EVs active): No action.

D.2 The 3600-Second Delay Bug

The v1 controller prototype contained a critical timing bug: outgoing HELICS endpoint messages were filtered through a 3600-second delay buffer intended for a different simulation mode. As a result, all controller shed commands arrived one hour after the overload was detected. In a 300-second experiment, no controller command ever took effect. This rendered the v1 controller non-functional during all v1-era experiments, producing artificially identical TVD values across all attacker variants (EVG = 1.0×). The v2 controller removes this delay entirely; commands take effect on the next HELICS time grant after the decision cycle.

D.3 Configuration Parameters

The configurable parameters are: `CTRL_INTERVAL_SEC` (observation interval, default 10 s), `CTRL_UPPER_THRESHOLD_W` ($P_{\max}$, default 4200 kW), `CTRL_LOWER_THRESHOLD_W` (lower threshold, default 2600 kW), `CTRL_RESTORE_HOLDOFF_SEC` (restore holdoff, default 30 s), and `CTRL_SEED` (shed-order random seed, default 42).

Table 4: Per-experiment raw data for all 37 completed experiments. Experiments are grouped by operating point and sorted by variant and seed.

Experiment	Variant	OP	Seed	Attacks	Unique EVs	TVD (s)
Hour 4 — Low Load (∼3480 kW, ∼720 kW headroom)						
random_hr4_s1	Random	Hr 4	1	3	2	60
random_hr4_s3	Random	Hr 4	3	3	2	0
random_hr4_s4	Random	Hr 4	4	3	2	60
random_hr4_s5	Random	Hr 4	5	3	2	120
ai_v1_hr4_s1	AI-V1	Hr 4	1	3	2	60
ai_v1_hr4_s2	AI-V1	Hr 4	2	3	2	60
ai_v1_hr4_s3	AI-V1	Hr 4	3	3	2	60
ai_v1_hr4_s4	AI-V1	Hr 4	4	3	1	0
ai_v1_hr4_s5	AI-V1	Hr 4	5	2	1	60
ai_v2_hr4_s1	AI-V2	Hr 4	1	4	4	180
ai_v2_hr4_s3	AI-V2	Hr 4	3	3	3	120
ai_v2_hr4_s4	AI-V2	Hr 4	4	4	4	120
ai_v2_hr4_s5	AI-V2	Hr 4	5	4	4	120
Hour 7 — Medium Load (∼3703 kW, ∼497 kW headroom)						
random_hr7_s1	Random	Hr 7	1	3	2	60
random_hr7_s3	Random	Hr 7	3	3	2	60
random_hr7_s4	Random	Hr 7	4	3	2	60
random_hr7_s5	Random	Hr 7	5	3	2	120
ai_v1_hr7_s1	AI-V1	Hr 7	1	3	1	120
ai_v1_hr7_s2	AI-V1	Hr 7	2	3	1	120
ai_v1_hr7_s3	AI-V1	Hr 7	3	3	1	120
ai_v1_hr7_s4	AI-V1	Hr 7	4	2	1	120
ai_v1_hr7_s5	AI-V1	Hr 7	5	3	1	120
ai_v2_hr7_s1	AI-V2	Hr 7	1	4	4	180
ai_v2_hr7_s2	AI-V2	Hr 7	2	4	4	180
ai_v2_hr7_s3	AI-V2	Hr 7	3	4	4	180
ai_v2_hr7_s4	AI-V2	Hr 7	4	4	4	180
Hour 14 — High Load (∼5490 kW, −1290 kW headroom, ceiling)						
random_hr14_s1	Random	Hr 14	1	3	2	295
random_hr14_s3	Random	Hr 14	3	3	2	295
random_hr14_s4	Random	Hr 14	4	3	2	295
ai_v1_hr14_s1	AI-V1	Hr 14	1	2	1	295
ai_v1_hr14_s2	AI-V1	Hr 14	2	3	1	295
ai_v1_hr14_s3	AI-V1	Hr 14	3	2	1	295
ai_v1_hr14_s4	AI-V1	Hr 14	4	2	1	295
ai_v1_hr14_s5	AI-V1	Hr 14	5	1	1	295
ai_v2_hr14_s1	AI-V2	Hr 14	1	4	4	295
ai_v2_hr14_s3	AI-V2	Hr 14	3	4	4	295
ai_v2_hr14_s5	AI-V2	Hr 14	5	4	4	295

Table 5: Hypothesis test results per operating point. All tests are one-sided Welch's t-tests ($\alpha = 0.05$). "∞" indicates zero within-group variance with different group means.

OP	Hypothesis	t	p	d	Means	Result
Hour 4 — Low Load						
Hr 4	H1: V2 TVD > Random TVD	2.61	0.020	1.85	135 vs. 60	**Reject**
Hr 4	H2: V2 TVD > V1 TVD	4.59	0.001	3.08	135 vs. 48	**Reject**
Hr 4	H3: V2 unique EVs > V1	6.07	<0.001	4.07	3.8 vs. 1.6	**Reject**
Hour 7 — Medium Load						
Hr 7	H1: V2 TVD > Random TVD	7.00	<0.001	4.95	180 vs. 75	**Reject**
Hr 7	H2: V2 TVD > V1 TVD	∞	<0.001	–	180 vs. 120	**Reject**
Hr 7	H3: V2 unique EVs > V1	∞	<0.001	–	4.0 vs. 1.0	**Reject**
Hr 7	H4: EVG > 1.0	∞	<0.001	–	EVG = 2.40	**Reject**
Hour 14 — High Load (Ceiling)						
Hr 14	H1: V2 TVD > Random TVD	–	–	0.00	295 vs. 295	Not sig. (ceiling)
Hr 14	H2: V2 TVD > V1 TVD	–	–	0.00	295 vs. 295	Not sig. (ceiling)
Hr 14	H3: V2 unique EVs > V1	∞	<0.001	–	4.0 vs. 1.0	**Reject**